Assignment 2: Extending the Python String Toolkit

Student ID	Student name	Assigned function	Example / expected output
62321	KAVYA CHAUDHARY	extract_sentences_with_number(text)	"Room 12 is ready. Welcome home." → ["Room 12 is ready."] Return sentences containing at least one digit.
62323	LAVISH MADAN	find_sentence_by_word_count(text, n)	"I am here. Hello there.", 2 → "Hello there." Return the first sentence containing exactly n words; return None if none match.
62352	TANAY JOSHI	remove_word(text, target)	"I like red apples", "red" → "I like apples" Remove the matching whole word and clean up extra spaces.
62356	AARNAV DEO	split_sentences(text)	"Hi there. How are you?" → ["Hi there.", "How are you?"] Split text into sentences while retaining sentence-ending punctuation.
62358	ABHISHEK BHARDWAJ	extract_email_addresses(text)	"Contact a@b.com or c@d.org" → ["a@b.com", "c@d.org"] Extract email-like strings using a basic pattern; no need to verify that they exist.
62359	ADEETYA KUNWAR SINGH	extract_acronyms(text)	"I study AI and ML" → ["AI", "ML"] Return uppercase tokens of at least two letters.
62364	ANSHIKA DHIMAN	extract_words_with_digits(text)	"room7 hall B2 lab" → ["room7", "B2"] Return whitespace-separated tokens containing at least one digit.
62368	ANUSHKA NEGI	normalize_whitespace(text)	"hello world\nPython" → "hello world Python" Replace runs of whitespace with a single space and trim the ends.
62373	AYUSH JOSHI	collapse_repeated_characters(text)	"heeellooo" → "helo" Reduce each run of consecutive identical characters to one character.
62374	AYUSH RAWAT	longest_sentence(text)	"I am here. Python is very useful." → "Python is very useful." Return the sentence containing the most words.
62385	KUSHAGRA SINGH	group_words_by_first_letter(text)	"apple ant ball" → {"a": ["apple", "ant"], "b": ["ball"]} Group words by their lowercase first letter.
62388	MEGHNA BAJETHA	generate_text_summary(text)	"Hello world. Hello Python." → dictionary with total words, unique words, and sentence count Return a dictionary with keys total_words, unique_words, and sentence_count.
62390	MONAAL BISHT	average_sentence_word_count(text)	"I like Python. It is useful." → 3.0 Return the average number of words per sentence.
62394	PAWAN RAWAT	shortest_sentence(text)	"I am here. Go now." → "Go now." Return the sentence containing the fewest words.
62398	PRAGYA SEMALTI	expand_repeated_characters(text)	"a3b2" → "aaabb" Interpret each letter followed by a digit as that many repetitions of the letter.
62401	RASHMI PANT	common_words(text1, text2)	"red blue green", "blue green yellow" → ["blue", "green"] Return shared unique words in the order they first appear in text1.
62403	SANIYA	compare_text_lengths(text1, text2)	"hello", "Python" → "second" Return "first", "second", or "equal" according to character count.
62404	SATYA NARAYAN PANDEY	unique_words(text)	"red blue red" → ["red", "blue"] Return each distinct word once, preserving first appearance.

62410	TANVI GOYAL	convert_snake_to_camel(text)	"hello_world" → "helloWorld" Convert snake_case text to camelCase.
62413	VANSH KAPIL	count_questions(text)	"How are you? I am fine. Why?" → 2 Count sentences ending in a question mark.
62415	VINAYAK MAHENDRU	remove_duplicate_words(text)	"red blue red green blue" → "red blue green" Keep the first occurrence of each word, preserving order.
62440	JIGYASHU BHATT	word_overlap_ratio(text1, text2)	"cat dog", "cat bird" → 1/3 Return Jaccard similarity: shared unique words divided by total unique words across both texts.
62475	ABHIJEET KRISHNA PANT	words_ending_with(text, suffix)	"walking talked running", "ing" → ["walking", "running"] Return words ending with the given suffix.
62477	AKSHAT MAITREY	remove_non_alphanumeric(text)	"hi@there! 123" → "hithere 123" Remove punctuation and symbols while retaining letters, digits, and spaces.
62483	ANSHIKA MAURYA	words_with_repeated_letters(text)	"book cat coffee" → ["book", "coffee"] Return words that contain at least one repeated letter.
62493	LOVELY BISHT	sentences_with_keyword(text, keyword)	"Python is easy. Java is popular.", "Python" → ["Python is easy."] Return sentences containing the keyword, ignoring case.
62497	MUKUL JOSHI	extract_uppercase_words(text)	"Python AI ML" → ["AI", "ML"] Return words written entirely in uppercase and containing at least one letter.
62500	PARTH VERMA	extract_mentions(text)	"Hello @sam and @riya" → ["@sam", "@riya"] Extract mentions beginning with @ and followed by letters, digits, or underscores.
62501	PARTIBHA SARKAR	convert_camel_to_snake(text)	"helloWorld" → "hello_world" Convert camelCase text to snake_case.
62502	PAYAL GUSAIN	truncate_text(text, limit)	"Hello Python World", 11 → "Hello Py..." Return text no longer than limit characters; use three dots when truncating.
62503	PRANAY GUPTA	is_valid_variable_name(text)	"student_name" → True Check Python identifier rules; keywords such as for and class must return False.
62504	RAKESH PRASAD	extract_words_in_quotes(text)	'Say "hello world" now' → ["hello world"] Extract text enclosed in matching double quotes.
62506	RASHMI DURGAPAL	reverse_each_word(text)	"hello world" → "olleh dlrow" Reverse each word while preserving word order.
62509	RISHIKA	count_substring_occurrences(text, sub)	"banana", "ana" → 2 Count overlapping occurrences of a non-empty substring.
62512	SAKSHAM GOEL	extract_hashtags(text)	"Learning #Python and #AI" → ["#Python", "#AI"] Extract hashtags beginning with # and followed by letters, digits, or underscores.
62518	UJWAL BHAN	sentence_word_counts(text)	"Hello there. I like Python." → [2, 3] Return the word count for each sentence in order.
62530	AYUSH SAKLANI	word_at_position(text, index)	"I like Python", 2 → "Python" Return the word at a zero-based index.
62540	HARSH GURURANI	sort_words_by_length(text)	"elephant cat dog" → ["cat", "dog", "elephant"] Return words from shortest to longest; preserve order for ties.
62544	HIMANSHU PALIWAL	insert_spaces_before_capitals(text)	"helloWorldPython" → "hello World Python" Insert a space before uppercase letters that follow lowercase letters.

62551	KARTIKAY PANT	find_all_substring_positions(text, sub)	"banana", "ana" → [1, 3] Return all starting positions of sub, including overlapping matches; use zero-based positions.
62557	MIHIR TIWARI	is_valid_email_format(text)	"student@example.com" → True Return True for a basic email-like format and False otherwise; this is not full email validation.
62563	PIYUSH SAINI	remove_digits_from_text(text)	"room12 block3" → "room block" Remove all digit characters.
62568	PULKIT SUCHDEVA	extract_urls(text)	"Visit https://example.com today" → ["https://example.com"] Extract URLs beginning with http:// or https://.
62574	SARTHAK KANDPAL	sort_words_alphabetically(text)	"pear apple mango" → ["apple", "mango", "pear"] Return the words in alphabetical order.
62842	PRATEEK POKHRIYAL	alternate_word_case(text)	"hello world python" → "HeLlO WoRlD PyThOn" Alternate uppercase and lowercase letters within each word, starting uppercase.
62859	DIVYANSHU SINGH	replace_word(text, old, new)	"cat and cat", "cat", "dog" → "dog and dog" Replace whole-word matches of old with new.
64033	KARAN SINGH	count_exclamations(text)	"Great! Hello. Amazing!" → 2 Count sentences ending in an exclamation mark.
64035	PAWAN SINGH	unique_words_in_first(text1, text2)	"red blue green", "blue yellow" → ["red", "green"] Return unique words that occur in text1 but not text2.
64037	SAURABH GUSAIN	mask_word(text, target)	"My password is secret", "secret" → "My password is ******" Replace each whole-word match with asterisks equal to the target word's length.
64040	ASTHA SHARMA	reverse_word_order(text)	"I like Python" → "Python like I" Reverse the order of words without reversing their letters.
64041	RIDHIMA NEGI	count_word_occurrences(text, word)	"hi bye hi", "hi" → 2 Count whole-word occurrences.
64042	SAKSHI	text_similarity_score(text1, text2)	"hello world", "hello world" → 1.0 Return a score from 0 to 1 using Jaccard similarity of unique words; identical word sets score 1.0.
64043	SIMRAN KAUR	words_starting_with(text, prefix)	"apple ant ball", "a" → ["apple", "ant"] Return words beginning with the given prefix.
64141	ABHISHEK YADAV	word_length_frequency(text)	"a to cat" → {1: 1, 2: 1, 3: 1} Return a dictionary mapping word length to the number of words of that length.